

When Does Adaptive Coordinate Sampling Pay for Itself?

Ridge-Regression Baselines and a Published ASCD Comparison

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Abstract

We study coordinate-selection rules for ridge regression under feature scaling and correlation. A baseline experiment compares uniform sampling, fixed Lipschitz sampling, periodically refreshed gradient-gain sampling, and a progress-triggered gradient-gain heuristic across four synthetic regimes and the packaged diabetes regression dataset. Each method is evaluated over 12 seeded runs using a common coordinate update and a 2,400-update budget. The two gradient-gain heuristics succeed on the independent synthetic designs but fail to reach the target on either correlated synthetic design within this budget.

As a subsequent published-method comparison, we implement the selection mechanism of Approximate Steepest Coordinate Descent (ASCD) of Stich, Raj, and Jaggi using their zero oracle with exact full-gradient initialization, while retaining the benchmark’s ridge coordinate update. ASCD reaches the target in all 12 runs on three of the four synthetic cases and on the diabetes dataset, and in 9 of 12 runs on the scaled-correlated case, compared with 6 of 12 for uniform sampling there. On the five scaled-correlated seeds where both methods reach the target, ASCD records greater proxy work on every seed under the current accounting convention. Its median active set also remains the full 48 coordinates on all four synthetic cases. The experiment therefore does not establish a general computational advantage for adaptive selection. It is a comparison on our benchmark rather than a reproduction of the published ASCD experiments, and no claim of algorithmic novelty or general superiority is made.

1 Motivation and related work

Randomized updates and adaptive information collection feature in stochastic block BFGS [1], minibatch semi-stochastic optimization [2], parallel and accelerated coordinate descent [3, 5], and randomized linear solvers [4]. Adaptive coordinate-selection methods include adaptive probabilities for stochastic dual coordinate ascent [6], adaptive importance sampling [7], and Approximate Steepest Coordinate Descent (ASCD) [8]. These works prevent gradient-informed sampling or refreshes alone from being treated as a new idea.

The practical question in the baseline experiment is whether the benefit of refreshing a sampling rule compensates for its additional computation under feature scaling and correlation. After completing that baseline, we additionally implement one published ASCD configuration and compare it under the same frozen benchmark. This second experiment is a cross-method comparison under our settings, not a reproduction of the published ASCD figures.

2 Experimental setup and baseline methods

For a design matrix $X \in \mathbb{R}^{n \times d}$ and response y , we minimize

$$f(w) = \frac{1}{2} \|Xw - y\|_2^2 + \frac{\lambda}{2} \|w\|_2^2, \quad \lambda = 0.03. \quad (1)$$

With residual $r = Xw - y$,

$$g_j = X_j^T r + \lambda w_j, \quad L_j = \|X_j\|_2^2 + \lambda, \quad (2)$$

and each baseline method applies

$$w_j \leftarrow w_j - \frac{g_j}{L_j} \quad (3)$$

and updates the residual.

Uniform sampling uses $p_j = 1/d$, whereas fixed Lipschitz sampling uses

$$p_j = \frac{L_j}{\sum_k L_k}.$$

Both gradient-gain variants use

$$p_j = \frac{0.05}{d} + 0.95 \frac{g_j^2/L_j}{\sum_k g_k^2/L_k}, \quad (4)$$

with uniform fallback when the denominator vanishes.

Scheduled gain refreshes initially and every 60 updates. Triggered gain refreshes at step zero and subsequently at a 120-step checkpoint if the latest objective decrease is below half the preceding decrease, subject to a 60-step minimum interval. The trigger observes the objective rather than the reference optimum.

The four synthetic cases cross balanced versus geometrically scaled features with independent versus eight-group correlated features, with $n = 400$ and $d = 48$. The packaged diabetes design provides a fifth case. Each method uses 12 problem seeds and at most 2,400 coordinate updates.

A dense ridge solve defines f^* for evaluation. A run succeeds if a saved checkpoint satisfies

$$f(w) - f^* \leq 10^{-4}.$$

The reference optimum is used only for evaluation and not by the selection rules.

For the baseline methods, the workload proxy counts $2n + 3$ units per coordinate update, d per coordinate draw, $nd + d$ per probability refresh, and $n + d$ per objective checkpoint. These units are bookkeeping quantities rather than measured floating-point-operation counts.

3 Baseline results

Table 1: Baseline runs reaching the target within the 2,400-update budget (out of 12).

Case	Uniform	Lipschitz	Scheduled	Triggered
Balanced independent	12	12	12	12
Scaled independent	12	0	12	12
Balanced correlated	12	12	0	0
Scaled correlated	6	0	0	0
Diabetes	12	12	9	5

Uniform and fixed Lipschitz sampling both succeed in all runs of the balanced-independent, balanced-correlated, and diabetes cases. Lipschitz sampling fails on the scaled-independent case, where the other three baseline methods succeed. The two gradient-gain heuristics fail to reach the target in either correlated synthetic case within the update budget.

Median time and work to target in the baseline summary use successful runs only. Consequently, such conditional summaries must be interpreted together with the corresponding success counts.

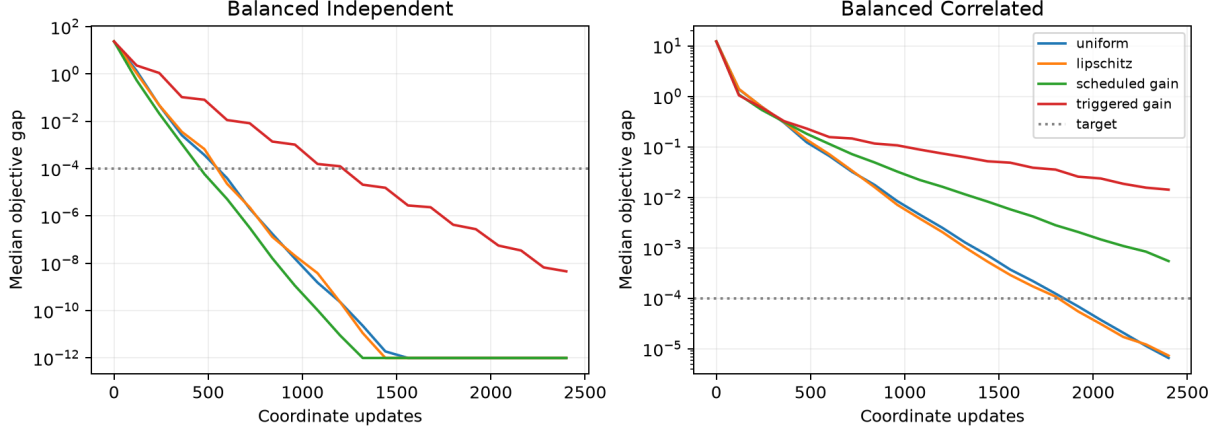


Figure 1: Median objective gaps for the two balanced baseline cases. The horizontal dotted line marks the 10^{-4} target. A median trajectory does not display the variability across individual runs.

4 Published ASCD comparison

4.1 Selection rule

We next implement a configuration of Approximate Steepest Coordinate Descent (ASCD) [8]. For every coordinate j , ASCD maintains an estimated gradient $\tilde{g}_j$ and a certified radius r_j satisfying

$$|g_j - \tilde{g}_j| \leq r_j.$$

This gives the upper and lower bounds

$$u_j = |\tilde{g}_j| + r_j, \quad \ell_j = \max\{0, |\tilde{g}_j| - r_j\}. \quad (5)$$

The active set is the smallest nonempty subset I for which every excluded coordinate satisfies

$$u_j^2 < \frac{1}{|I|} \sum_{i \in I} \ell_i^2. \quad (6)$$

Equality therefore does not permit exclusion. A coordinate is selected uniformly from the members of I attaining the largest lower bound.

Our implementation initializes $\tilde{g}$ using the exact full gradient and sets all radii to zero. This is an allowed ASCD configuration discussed by Stich et al.; it differs from the default zero-estimate, infinite-radius initialization in their Algorithm 1.

4.2 Zero oracle and radius update

For this experiment we use the paper’s zero oracle for cross-coordinate gradient changes. If coordinate i changes by h , then for $j \neq i$ the ridge gradient changes by

$$g_j^{\text{new}} - g_j = h X_j^\top X_i. \quad (7)$$

Cauchy–Schwarz gives

$$|g_j^{\text{new}} - g_j| \leq |h| \|X_j\|_2 \|X_i\|_2. \quad (8)$$

The implementation therefore leaves the corresponding estimated cross-gradient unchanged and increases its certified radius by this upper bound.

For the selected coordinate,

$$h = -\frac{g_i}{L_i},$$

so exact minimization along that coordinate gives

$$g_i^{\text{new}} = g_i + hL_i = 0$$

up to numerical precision. Its estimate and radius can therefore be reset to zero.

4.3 Implementation verification

The implementation includes a small-problem numerical audit. After every audited update, it recomputes

$$X^\top(Xw - y) + \lambda w$$

and verifies that each true coordinate gradient lies inside its maintained interval. It also verifies the residual invariant $r = Xw - y$.

A manual active-set trace gives the same result as the implementation. For example, with exact estimates

$$\tilde{g} = (3, 1, 0), \quad r = (0, 0, 0),$$

the initial mean squared lower bound is $10/3$. The zero coordinate can therefore be excluded. For the remaining two coordinates the mean is 5, so the coordinate with squared upper bound 1 can also be excluded, leaving only the coordinate with gradient magnitude 3.

Two useful additional automated audit checks are an explicit equality-boundary test for the strict exclusion condition and an assertion that the objective is nonincreasing after each audited exact coordinate minimization.

4.4 ASCD results

The ASCD comparison uses the same five cases, 12 problem seeds, initial point, objective, 10^{-4} target, 120-update checkpoint spacing, and 2,400-update limit as the baseline.

Table 2: Uniform and ASCD results on the frozen benchmark. Work to target is the median among successful runs under each method’s recorded proxy.

Case	Uniform		ASCD		
	Success	Work	Success	Work	Median $ I $
Balanced independent	12/12	513,288	12/12	762,936	48
Scaled independent	12/12	513,288	12/12	762,936	48
Balanced correlated	12/12	1,641,536	12/12	2,249,416	48
Scaled correlated	6/12	1,897,956	9/12	2,844,008	48
Diabetes	12/12	919,234	12/12	984,864	10

ASCD reaches the target in all 12 runs on the balanced-independent, scaled-independent, balanced-correlated, and diabetes cases. On the scaled-correlated case it reaches the target in 9 of 12 runs, compared with 6 of 12 for uniform sampling.

The work-to-target medians in Table 2 are conditional on successful runs. In particular, the scaled-correlated uniform and ASCD medians are calculated from different sets of successful problem seeds, so those two aggregate values alone are not a paired cost comparison.

The active-set measurements are also informative. The median active set is 48, the full coordinate dimension, on all four synthetic cases. It is 10 on the 10-feature diabetes case. Thus this zero-oracle configuration shows little active-set pruning under the synthetic settings. The relatively broad certified Cauchy–Schwarz bounds are a plausible mechanism, but a direct per-step analysis would be required to establish the cause.

4.5 Paired scaled-correlated comparison

Because the uniform and ASCD successful-run medians in Table 2 are calculated from different subsets of problem seeds, we additionally compare the methods seed by seed on the scaled-correlated case. For each method and seed, work to target is the proxy work recorded at the first saved checkpoint satisfying $f(w) - f^* \leq 10^{-4}$.

Among the 12 paired problem seeds, uniform reaches the target on 6 seeds, whereas ASCD reaches it on 9. The two methods both reach the target on five seeds: 0, 4, 5, 6, and 10. On these shared successes, the median recorded work to target is 1,846,672 for uniform and 2,844,008 for ASCD. The median paired ASCD-minus-uniform difference is 997,336 proxy units, corresponding to a median ASCD/uniform recorded-work ratio of 1.5401. ASCD records greater proxy work on all five shared successes.

The paired runs also distinguish success-set differences. Seed 3 reaches the target under uniform sampling but not under ASCD, whereas seeds 1, 7, 8, and 9 reach the target under ASCD but not under uniform sampling. Thus the higher ASCD success count is not obtained simply by reproducing the same successful instances.

This paired result separates two observations. Under the fixed 2,400-update budget, ASCD succeeds on more scaled-correlated problem instances. On instances where both methods succeed, however, the current proxy accounting does not show a lower work-to-target cost for ASCD. The median recorded ASCD work on the shared successes is about 1.54 times the corresponding uniform median under the current accounting conventions. These are descriptive results under the present cost model; because setup and accounting boundaries are not fully harmonized, they should not be interpreted as exact operation-count or wall-clock superiority claims.

4.6 Work accounting

ASCD includes an initial full-gradient charge of

$$nd + d$$

proxy units. Each saved objective checkpoint adds $n + d$. Each coordinate update adds

$$2n + 3 + d\lceil \log_2 d \rceil + 3d,$$

representing the coordinate update, modelled sorting, and bounds/interval maintenance.

These quantities are deliberately approximate bookkeeping units. Some common matrix construction and data-generation work occurs outside the timed ASCD method section, while ASCD’s full-gradient initialization is inside it. The baseline and ASCD timing/setup boundaries are therefore not sufficiently harmonized to support a controlled wall-clock or exact operation-count superiority claim.

The paired comparison above therefore uses the proxies exactly as recorded by the two implementations, but it does not remove the difference in accounting conventions. The resulting work differences are descriptive under this cost model rather than controlled FLOP or wall-clock comparisons.

5 Interpretation and limitations

The baseline results show that neither gradient-gain heuristic has a consistent advantage over the fixed rules. In particular, both fail to reach the target in the correlated synthetic regimes within the 2,400-update budget.

The published ASCD comparison changes the picture but does not establish a general computational advantage. ASCD reaches the target in more scaled-correlated runs than uniform

sampling under the fixed update budget. The paired analysis shows that, on the five scaled-correlated instances where both methods succeed, ASCD records greater proxy work on every instance under the present accounting convention. Moreover, the median active set remains the entire coordinate set on all synthetic cases. Under this zero-oracle configuration, the additional machinery therefore does not translate into substantial observed pruning on these problems.

The comparison has several limitations. Only one small packaged real dataset is included, and no predictive generalization performance is evaluated. The baseline trigger threshold, refresh interval, and probability floor are illustrative choices. Proxy work is an accounting measure rather than an exact FLOP count, and wall-clock measurements depend on both implementation and hardware. Objective gaps reported as zero can reflect floating-point precision.

The ASCD experiment should also not be described as a reproduction of the figures in Stich et al. Their ridge-regression experiments use different synthetic settings and oracle configurations. A strict reproduction would require the corresponding data generation, initialization, oracle precision, step-size/update convention, and plotting protocol.

Finally, the paper’s theoretical one-step comparison assumes a shared coordinate smoothness bound and a common update method across the algorithms being compared. The present benchmark uses coordinate-specific L_j values and its existing exact coordinate-minimization update. The paper’s dominance result therefore should not be interpreted as a theoretical guarantee for the experimental comparison reported here.

6 Conclusion and future research

The baseline study and an initial published-method comparison are now complete. The baseline gradient-gain heuristics do not succeed consistently across the five cases, with the correlated synthetic regimes providing a useful failure case. The subsequent ASCD implementation reaches the scaled-correlated target in 9 of 12 runs under the same update budget, compared with 6 of 12 for uniform sampling.

The paired comparison sharpens this result. Five scaled-correlated seeds succeed under both methods, and ASCD records greater proxy work on each of those five. Median work to target on the shared successes is 1,846,672 for uniform and 2,844,008 for ASCD, a median recorded-work ratio of 1.5401. Together with the full-size median active sets on the synthetic cases, this means that the current zero-oracle ASCD configuration does not demonstrate a cost advantage under the present work model, despite its higher success count under the fixed coordinate-update budget.

The next useful step is therefore not to claim superiority or novelty, but to sharpen the remaining cost and mechanism analysis. The paired per-seed scaled-correlated comparison is now complete; remaining work includes harmonizing setup and cost accounting and examining the distribution of active-set sizes over individual updates. A subsequent predeclared experiment could evaluate a tighter *published* ASCD oracle, such as an approximate inner-product oracle, while explicitly charging its information-acquisition cost. Broader datasets, held-out experimental settings, and repeated wall-clock measurements would be needed before drawing stronger conclusions.

Any later methodological modification should be separated clearly from the published-method comparison and evaluated under a predeclared question and evaluation protocol. The internship research topic would be determined with the faculty member if an application succeeds.

Acknowledgement and research provenance. AI assistance was used in developing the initial benchmark, the ASCD implementation and audit, the paired-analysis workflow, and report text. The code and generated output files accompany the report for reproducibility. The ASCD

active-set condition and gradient-radius derivation were manually reviewed during Stage 3; descriptions of independent authorship should reflect the author’s actual personal verification, decisions, and revisions. Applicable citation and AI-disclosure requirements should be followed.

References

- [1] R. M. Gower, D. Goldfarb, and P. Richtárik. Stochastic block BFGS: Squeezing more curvature out of data. *Proceedings of ICML*, 2016. <https://proceedings.mlr.press/v48/gower16.pdf>.
- [2] J. Konečný, J. Liu, P. Richtárik, and M. Takáč. Mini-batch semi-stochastic gradient descent in the proximal setting. *IEEE Journal of Selected Topics in Signal Processing*, 2016. <https://www.pure.ed.ac.uk/ws/portalfiles/portal/26531306/07347336.pdf>.
- [3] P. Richtárik and M. Takáč. Parallel coordinate descent methods for big data optimization. *Mathematical Programming*, 2016. <https://arxiv.org/abs/1212.0873>.
- [4] R. M. Gower and P. Richtárik. Randomized iterative methods for linear systems. *SIAM Journal on Matrix Analysis and Applications*, 2015. <https://arxiv.org/abs/1506.03296>.
- [5] O. Fercoq and P. Richtárik. Accelerated, parallel, and proximal coordinate descent. *SIAM Journal on Optimization*, 2015. <https://arxiv.org/abs/1312.5799>.
- [6] D. Csiba, Z. Qu, and P. Richtárik. Stochastic dual coordinate ascent with adaptive probabilities. <https://arxiv.org/abs/1502.08053>.
- [7] D. Perekrestenko, V. Cevher, and M. Jaggi. Faster coordinate descent via adaptive importance sampling. *Proceedings of AISTATS*, 2017. <https://proceedings.mlr.press/v54/perekrestenko17a.html>.
- [8] S. U. Stich, A. Raj, and M. Jaggi. Approximate steepest coordinate descent. *Proceedings of ICML*, 2017. <https://proceedings.mlr.press/v70/stich17a.html>.